

Kunj Rathod

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Software engineer focused on native Apple platform development and privacy-first, on-device intelligence: Swift/SwiftUI apps and Rust systems with Metal-accelerated inference on Apple Silicon.

Education

University of Utah

B.S. in Computer Science | GPA: 3.7/4.0, Dean's List

Aug 2023 – Dec 2026

Salt Lake City, UT

Coursework: Operating Systems, Distributed Systems, Computer Vision, Machine Learning, NLP, Algorithms, Databases.

Apple Platform & On-Device Projects

FNDR – Privacy-First Local AI Assistant for macOS

Jan 2024 – Present

Rust, Metal, Apple Vision Framework, ONNX, Llama 3.2, Whisper

- Built a zero-trust, local-only memory assistant for macOS with full data sovereignty: no cloud, no telemetry, all inference and storage on device.
- Optimized on-device LLM (Llama 3.2) and VLM (SmolVLM) inference with Metal-accelerated backends for low-latency RAG on M-series Apple Silicon.
- Architected a real-time screen-extraction pipeline on the Apple Vision Framework for high-speed OCR and CLIP visual embeddings, reconstructing temporal context from screen snapshots.
- Added on-device meeting intelligence with local Whisper transcription and a temporal search engine over vector + lexical fields, modeling semantic links across activity, web, and audio.

Wingman – Multi-Modal AI Personal Assistant (iOS)

Apr 2025 – Present

SwiftUI, MVVM, OpenAI API (GPT-4o, Whisper), Firebase

- Built a native iOS assistant in SwiftUI on a clean MVVM architecture, supporting three input modalities in one conversational surface: voice, text chat, and image.
- Integrated GPT-4o for multimodal, context-aware responses and Whisper for voice transcription, grounded by RAG-enhanced memory over prior conversation history.
- Implemented an offline-first design with Firebase sync, real-time streamed message rendering, and persistent conversation history across sessions.

Experience

Software Engineering Intern – Fabric (Azure Data)

May 2026 – Aug 2026

Microsoft

Redmond, WA

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal, with telemetry-backed auditable change timelines across large enterprise tenants.
- Defining data contracts with UX and backend teams across 3 time zones; migrating capacity workflows into native platform UX.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health

Salt Lake City, UT

- Architected serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms time-to-first-token and 40% latency reduction via pipeline optimization and caching.
- Owned API design, schema, and cloud infrastructure; shipped 6 features across 4 sprints with distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen

Remote

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents (+28% recall, -35% hallucinations); benchmarked 8 model families and built routing that cut inference spend.

Research

Mechanistic Interpretability & Embodied Agents

2026 – Present

University of Utah

- MEMROT: reproducible pipeline probing long-context memory failure in Gemma-2-2B via SAE layer scans and mean-ablation interventions on distributed compute.
- Spatial memory for embodied agents: retrieval-augmented spatial memory for long-horizon navigation; evaluated on OS Marathon, BrowseComp, and Mind2Web.

Technical Skills

Apple / Native: Swift, SwiftUI, MVVM, Xcode, Metal, Apple Vision Framework, on-device LLM/VLM inference, Apple Silicon

Languages: Swift, Rust, C++, Python, TypeScript, Java, SQL

Systems & ML: ONNX, PyTorch, RAG/GraphRAG, LanceDB, MCP, FastAPI, Docker, AWS (Bedrock, Lambda), CI/CD